Huawei Access Node HEB-ACC-75 Optical Transport Failure SOP

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2641

Description

This Standard Operating Procedure (SOP) addresses the failure of the Optical Transport subsystem within the Huawei Access Node HEB-ACC-75 at the Hebbal Edge site. The failure is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss, which falls under the scope of the Access role. The Optical Transport Failure is a critical component of the overall network architecture, as it enables high-speed, long-distance transmission of optical signals between nodes. In the Huawei equipment, the Optical Transport subsystem utilizes the Optical Transport Network (OTN) protocol, which is responsible for framing, multiplexing, and managing the optical signals. The OTN protocol is built on top of the Synchronous Digital Hierarchy (SDH) or Synchronous Optical Networking (SONET) protocols, which provide the underlying framework for signal transmission. The failure of the Optical Transport subsystem can lead to service disruptions, packet loss, and increased latency, ultimately impacting the overall network performance and user experience. Therefore, timely identification and resolution of Optical Transport failures are crucial to maintaining network uptime and reliability.

Pre-requisite Checks

- Check: Verify the site location and ensure that the node is accessible for remote hands
- Check: Confirm the redundant link status and ensure that it is operational
- Check: Gather equipment information, including the node ID, site location, and vendor CLI dialect

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cut or damage	Medium
Rx/Tx power loss due to optical signal degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport link verbose, Expected Output: Link ID: 1, Status: Up, Tx Power: 0 dBm, Rx Power: -3 dBm, Description: This command displays detailed information about the optical link, including its status, Tx power, and Rx power. The expected output indicates that the link is up, but the Rx power is lower than the expected value, suggesting potential signal degradation.

Command: display optical-transport sfp verbose, Expected Output: SFP ID: 1, Status: Ok, Type: 100G SR4, Rx Power: -3 dBm, Tx Power: 0 dBm,
```

```
Description: This command displays detailed information about the SFP transceiver, including its status, type, Rx power, and Tx power. The expected output indicates that the SFP is operational, but the Rx power is lower than the expected value, suggesting potential degradation.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace optical-transport sfp 1 with , Description: This step involves replacing the degraded SFP/XFP transceiver with a new one. The command 'replace optical-transport sfp 1 with ' is used to replace the SFP with a new one, where is the ID of the new SFP.

Step 2: Step: Repair or replace the fiber, Command: repair optical-transport fiber 1, Description: This step involves repairing or replacing the damaged fiber. The command 'repair optical-transport fiber 1' is used to repair or replace the fiber, where 1 is the ID of the fiber to be repaired or replaced.

Post-Check Verification

```
Command: display optical-transport link verbose, Expected Output: Link ID: 1, Status: Up, Tx Power: 0 dBm, Rx Power: -3 dBm, Description: This command is used to verify that the optical link is operational and the Rx power is within the expected range.

Command: display optical-transport sfp verbose, Expected Output: SFP ID: 1, Status: Ok, Type: 100G SR4, Rx Power: -3 dBm, Tx Power: 0 dBm, Description: This command is used to verify that the SFP transceiver is operational and the Rx power is within the expected range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore optical-transport sfp 1 to original, Description: This step involves restoring the original SFP/XFP transceiver. The command 'restore optical-transport sfp 1 to original' is used to restore the SFP to its original state.

- Step: Restore the original fiber, Command: restore optical-transport fiber 1 to original, Description: This step involves restoring the original fiber. The command 'restore optical-transport fiber 1 to original' is used to restore the fiber to its original state.